WATERMARK FORGING FOR TRUSTWORTHY IMAGE AUTHENTICATION
TRUSTWORTHY MACHINE LEARNING

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1 INTRODUCTION

In this assignment, the goal is to perform a watermark forging attack under black-box conditions. We are given 200 clean target images and 8 groups of watermarked source images. Every group contains the same hidden watermark, but we do not know which watermarking method is used. The main goal is to copy the watermark from the source images and apply it to the clean target images without changing the image quality too much. The final score depends on both watermark detection accuracy and visual quality, so both are equally important.

2 APPROACH

Our final solution combines different watermark extraction methods together with known watermark encoders whenever the watermarking scheme could be identified.

First, we used Non-Local Means (NLM) denoising to remove the image content and get the residual noise from every source image. Since all 25 images in one group contain the same hidden watermark, we averaged these residuals using Phase Locking Value (PLV) in the Fourier domain. This helps remove random image noise and keeps only the common watermark signal.

Then we also used a preference model based extraction method. We used a pretrained ConvNeXt model and performed gradient optimization to estimate the hidden watermark pattern from the source images. This method gives another watermark template which captures different watermark information.

Instead of using only one method, we combined both templates because both performed well in different situations. The final watermark template was created using:

35% Preference Model + 65% NLM + PLV

This combination gave better leaderboard results than using only one extraction method. For embedding the watermark into the clean target images, we converted the watermark into YCbCr color space. We increased the embedding strength mainly in the chroma channels because they affect image quality less than the luminance channel. After that, we applied a JND (Just Noticeable Difference) texture mask so that stronger watermark is added only in textured regions and weaker watermark is added in smooth areas. Finally, we used binary search to find the maximum embedding strength while keeping the LPIPS value inside the allowed quality budget.

During our experiments, we were also able to identify some watermarking methods directly. Instead of estimating their watermark, we used their original encoder.

The identified watermark schemes are:

WM_2 $\rightarrow$ RivaGAN

WM_4 $\rightarrow$ VINE

WM_7 $\rightarrow$ TrustMark

Using the original encoder gave much better watermark detection while still keeping very good image quality.

Table 1: Main settings and parameters used in the final solution.

Parameter	**Value**
Preference template	35%
NLM + PLV template	65%
LPIPS budget	0.022
Preference extraction steps	400
Preference model	ConvNeXt
Watermark extraction	NLM + PLV + Preference Model
Embedding	YCbCr + JND Mask
Known scheme override	RivaGAN, VINE, TrustMark

Finally, we generated a ZIP file containing all 200 forged images according to the required mapping and submitted it to the evaluation server.

3 Public Leaderboard Result

For the final submission, we generated a ZIP file containing all the 200 forged images with the copied watermarks.

Our final method is:

- Hybrid Watermark Forgery
- NLM + PLV watermark extraction
- Preference model extraction
- 35/65 blended watermark template
- YCbCr based embedding
- JND texture masking
- LPIPS constrained embedding
- Known watermark encoder overrides

My best public leaderboard result was:

```
Public leaderboard score:  0.781984
```

This score was achieved by combining watermark estimation with the original encoders for the identified watermarking methods. Compared to using only one extraction method, this hybrid approach improved watermark detection while still maintaining very good visual quality.

4 Conclusion

From this assignment, I understood that even if the watermarking algorithm is hidden, it is still possible to forge watermarks by using different extraction methods and carefully embedding them into new images. Combining NLM, preference model extraction, and known watermark encoders helped improve both watermark detection and image quality. This also shows that current watermarking systems still have security limitations, and stronger watermarking methods are needed to defend against watermark forgery attacks. Overall, our final solution uses a simple but effective pipeline consisting of NLM residual extraction, preference model extraction, blended watermark templates, YCbCr embedding, JND masking, and known watermark encoder overrides. This gave a good balance between watermark recovery and visual quality for this assignment.

GitHub Repository: `https://github.com/SanjaySSrivathsa/Assignment4_TML`